

FGF21 agonists: An emerging therapeutic for metabolic dysfunction-associated steatohepatitis and beyond

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Summary

The worldwide epidemics of obesity, hypertriglyceridemia, dyslipidaemia, type 2 diabetes, and metabolic dysfunction-associated steatotic liver disease (MASLD)/metabolic dysfunction-associated steatohepatitis (MASH) represent a major economic burden on healthcare systems. Patients with at-risk MASH, defined as MASH with moderate or significant fibrosis, are at higher risk of comorbidity/mortality, with a significant risk of cardiovascular diseases and/or major adverse liver outcomes. Despite a high unmet medical need, there is only one drug approved for MASH. Several drug candidates have reached the phase III development stage and could lead to several potential conditional drug approvals in the coming years. Within the armamentarium of future treatment options, FGF21 analogues hold an interesting position thanks to their pleiotropic effects in addition to their significant effect on both MASH resolution and fibrosis improvement. In this review, we summarise preclinical and clinical data from FGF21 analogues for MASH and explore additional potential therapeutic indications.

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Introduction

Fibroblast growth factor (FGF) 21 is a stress-induced 181 amino acid hormone produced in the liver and other metabolically active tissues. FGF21 is primarily secreted from the liver and adipose tissue, which regulates energy homeostasis and has pleiotropic metabolic effects. The secretion of FGF21 is induced by several extrinsic stressors such as exercise, fasting, protein deficiency, high fructose intake, cold exposure, and alcohol consumption.^{1–6} FGF21's biological actions are mediated by binding to fibroblast growth factor receptors (FGFRs) 1c, 2c and 3c thereby eliciting multiple intracellular signalling pathways. The coreceptor β -klotho, which serves as a high affinity trap for FGF21, is essential since it forms a trimeric coreceptor-FGF21 complex necessary for signalling via FGFRs.^{7,8} Native FGF21 notably targets adipose tissue to increase glucose uptake,⁹ adiponectin secretion¹⁰ and regulate lipid metabolism.¹¹ Several FGF21 analogues are currently in development and have been studied in numerous metabolic disease indications such as obesity, hypertriglyceridemia, dyslipidaemia, type 2 diabetes, and metabolic dysfunction-associated steatotic liver disease (MASLD) and its more advanced form, metabolic dysfunction-associated steatohepatitis (MASH).^{12–21} No FGF21 analogues have been approved yet by the US Food and Drug Administration (FDA) or the European Medicines Agency (EMA). The two most advanced drug candidates for this class of therapeutics, efruxifermin and pegozafermin, have met their phase IIb primary and secondary endpoints in patients with MASH. Phase III programs have been initiated for both

efruxifermin and pegozafermin to confirm the potential of FGF21 analogues as treatments for patients with MASH and fibrosis. This review covers the clinical rationale for therapies based on FGF21, and the available clinical data for FGF21 analogues in different disease indications, with a particular focus on MASH (Fig. 1).

FGF-21 background

Introduction to FGF21

FGF21 is a member of the endocrine sub-family of the FGF super-family, closely related to FGF19 and FGF23.²² While members of this sub-family interact with the same FGFRs as canonical FGFs, they are distinguished by depending on either α - (FGF23) or β -klotho (FGF19 and FGF21) as a coreceptor rather than hyaluronic acid. One of the purposes of the coreceptor is to serve as a trap for FGFs on the cell surface. By not binding to hyaluronic acid, the endocrine, unlike canonical FGFs, are able to escape from the immediate interstitial environment they are secreted into, allowing them to enter the systemic circulation and act as endocrine hormones.²³ Their target tissues are limited to those which express the coreceptor β -klotho. In humans, the major tissues or organs regulated by the endocrine actions of FGF21 appear to be the liver, adipose tissue (white and brown), pancreas, small-intestine and muscle which all express β -klotho.²⁴

Like canonical FGFs, FGF21 also exerts autocrine and paracrine effects on cells in which β -klotho is expressed.

Keywords: MASLD; MASH; liver fibrosis; FGF21 analogs.

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Keypoints

- FGF21 analogues possess a “directly” acting antifibrotic mechanism of action, reducing fibrogenesis rapidly after treatment starts, with significant regression of fibrosis after 24 weeks’ treatment.
- Concurrently FGF21 analogues reduce liver steatosis while activating intra-cellular pathways to protect hepatocytes against stress, thereby reducing the underlying driver of liver injury, which with longer-term treatment “indirectly” reduces fibrogenesis.
- By also improving metabolic health of the whole body, increasing insulin sensitivity, and reducing dyslipidemia, FGF21 analogues are expected to sustainably improve liver health, resulting in further regression of fibrosis with longer term treatment.
- Preliminary observations suggest that when used with GLP-1 receptor agonists, FGF21 analogues bring additional reductions in liver injury and fibrogenesis, with their insulin-sensitizing property complementing the secretion of insulin by GLP-1 receptor agonists.
- FGF21 analogues appear generally well tolerated with the most frequent side effects being transient, mild or moderate and gastrointestinal in origin.

Physiological role of FGF21 in the context of MASH

A major role of FGF21 during human evolution appears to have been to protect the livers of food gathering ancestors when stressed by excessive ingestion of fructose and alcohol associated with binge eating of (often partially fermented) fruit.²⁵

There was a survival benefit of being able to consume a large amount of energy quickly to avoid falling to predators, while also sustaining life until the next meal of unpredictable availability. However, this placed major stress on the liver because the metabolism of fructose is an essentially unlimited process

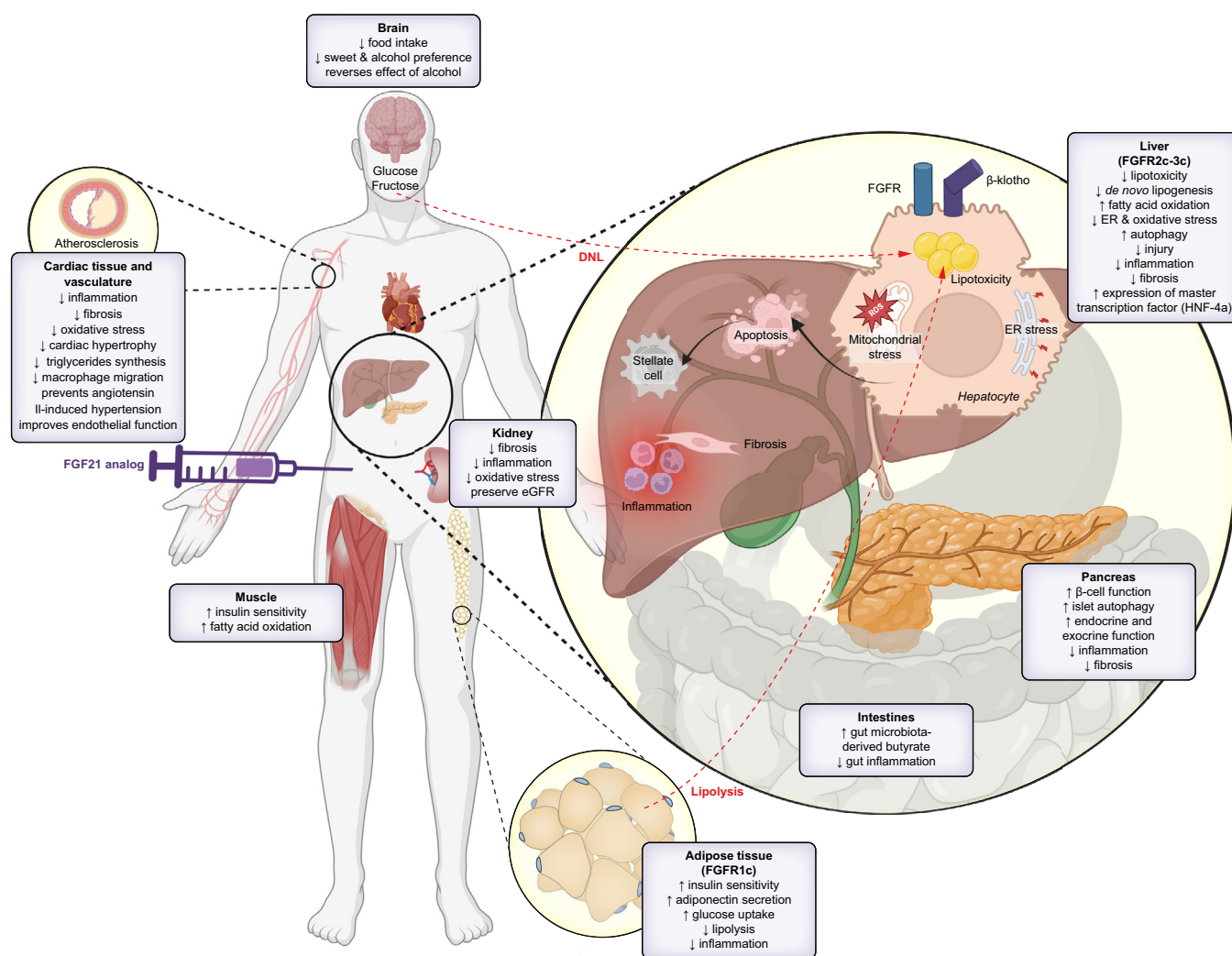


Fig. 1. Mechanism of Action of FGF21 analogues. FGF, fibroblast growth factor; FGFR, fibroblast growth factor receptor.

that rapidly consumes a lot of ATP.²⁵ This stress is associated with rapid induction of expression of FGF21 by hepatocytes.³ FGF21 activates many pathways via its endocrine, autocrine and paracrine effects to alleviate this stress and protect hepatocytes.

As a further aid to survival, FGF21 is also induced by uptake and metabolism of alcohol by hepatocytes.⁴ This serves not only to increase the capacity to metabolise alcohol, but also alleviates intoxication with excessive alcohol, by activating noradrenergic centres in the brain via the dorsal vagal complex.²⁶ FGF21 also reduces the desire to ingest alcohol.⁴

Endocrine effects

FGF21 appears to play a major role in balancing the intake of different classes of macronutrient. Levels of FGF21 increase when the content of protein in the diet is low,^{27,28} or when levels of fructose or carbohydrate are high.³ In the latter instance, reducing the desire to eat sweet-tasting foods.^{29–31} FGF21 is an important determinant of insulin sensitivity. By enhancing sensitivity to insulin under conditions of positive caloric balance,^{15,32} FGF21 promotes energy storage particularly in peripheral adipose tissue and skeletal muscle. Under conditions when caloric intake greatly exceeds demand, FGF21 serves to redirect energy away from the liver, an important aspect of FGF21's endocrine effects which can be leveraged to treat MASLD/MASH. Indeed, FGF21 increases insulin sensitivity of peripheral tissues, notably adipose tissue. As a result, insulin is better able to inhibit the release of free fatty acids from adipose tissue into the circulation; these free fatty acids are ultimately taken up by hepatocytes. Insulin is also better able to promote uptake of chylomicrons containing dietary fat from the gastrointestinal tract, and of very low-density lipoprotein secreted by hepatocytes containing triglycerides (TGs) synthesised *de novo* from carbohydrate (fructose in particular) and branched-chain amino acid carbon.

Autocrine effects

FGF21 expression appears to be induced by metabolic dysfunction in any organ or tissue. For example, systemic levels are higher in individuals with rare mitochondrial disorders resulting in chronically high levels of oxidative and endoplasmic reticulum (ER) stress, both in muscle and the nervous system.^{33,34} Systemic levels are also elevated in patients who have type 2 diabetes, MASH and obesity,^{35–37} and in patients with cardiovascular³⁸ and renal disease. Oxidative stress induced by chemical injury (bleomycin in lung; dimethylnitrosamine,³⁹ thioacetamide,⁴⁰ ethanol and acetaminophen in hepatocytes) also induces expression of FGF21.

FGF21 protects against ER stress e.g. induced in hepatocytes by diets deficient in methionine and choline⁴¹ or acute episodes of viral hepatitis,⁴² or in pancreatic acinar cells by high rates of protein synthesis.⁴³

FGF21 modulates a number of intracellular pathways to protect hepatocytes against stress and ultimately cell death.⁴⁴ For example, it increases the capacity of antioxidant pathways via activation of the transcription factor Nrf2.⁴⁵ Likewise, it increases autophagy pathways, including lysosomal biogenesis, via activation of transcription factor EB,^{46,47} to degrade

damaged or misfolded proteins underlying ER stress, and by suppressing induction of pro-apoptotic pathways.

Through its autocrine action, FGF21 may also more broadly influence the phenotype of hepatocytes. It has been shown to increase expression of the master transcription factor HNF-4 α (hepatocyte nuclear factor-4 α), which regulates the functional capacity of hepatocytes including albumin synthesis, gluconeogenesis, lipid metabolism and cytochrome P450-mediated oxidative pathways.⁴⁸ Expression of HNF4 α is reduced in hepatocytes of patients with MASH.⁴⁹

Elevated levels of FGF21 associated with chronic metabolic stress or dysfunction in humans,^{35,50,51} including patients with MASLD/MASH, has led to the hypothesis that resistance to endogenous FGF21 develops.^{36,52,53} It has been reported that levels of β -klotho expression are lower in tissues of individuals with chronic metabolic dysfunction, which could underlie this resistance.⁵⁴

Paracrine effects

FGF21 released by stressed hepatocytes also appears to modulate the activity and phenotype of adjacent Kupffer cells and hepatic stellate cells (HSCs) in the liver.

In vitro, FGF21 appears to shift the phenotype of macrophages from a pro-inflammatory to a pro-repair phenotype.⁵⁵ Treatment of non-human primates with an FGF21 analogue was associated with reduced expression of pro-inflammatory markers and increased expression of anti-inflammatory markers.⁵⁶

In vitro, FGF21 has been shown to directly suppress differentiation of HSCs into myofibroblasts.^{57,58} In preclinical *in vivo* models of fibrosis induced by chemical agents or pathophysiological levels of stress, loss of FGF21 has been shown to exacerbate development of fibrosis in the lungs, pancreas, liver, kidney and heart, while its restoration or over-expression have been shown to mitigate fibrosis.^{59–64} Human genotype-phenotype association studies have revealed that the loss of function of the FGF21 pathway appears to be associated with increased potential for fibrosis in the liver.^{65–67}

In conclusion, FGF21 analogues appear to act on all stages of the cascade of pathogenesis underlying MASH, clearing lipids from and protecting hepatocytes from stress, suppressing apoptosis and pro-inflammatory signalling, and inhibiting differentiation of HSCs into collagen-secreting myofibroblasts.

Crosstalk with other hormones

In the context of MASH, FGF21 modulates the activity of other hormones, besides insulin, involved in regulating whole body and liver metabolism. It does this by acting at two levels: by altering expression of their receptors, or by changing levels of the hormones themselves. For example, FGF21 has been reported to reduce glucagon levels in non-human primates, although levels appeared unchanged or slightly increased in humans,⁶⁸ while it appears to increase expression of the glucagon receptor in the liver. FGF21 appears to potentiate the actions of thyroid hormone in the liver by increasing the activity of deiodinase-1 and upregulating expression of thyroid hormone receptor β (THR- β). On

balance, these changes to glucagon and thyroid hormone signalling elicited by FGF21 would be expected to increase rates of fatty acid oxidation and reduce triglyceride synthesis and deposition in the liver.⁶⁹

FGF21 also stimulates secretion of the adipokine adiponectin from adipose tissue, which may mediate or contribute to FGF21's metabolic, anti-inflammatory and antifibrotic effects. Interpretation of the biology of adiponectin is complicated by its existence as four discrete forms: globular, or low (trimer), medium (hexamer) or high molecular weight (multimer). For example, the high molecular weight form can be pro-inflammatory, e.g. in rheumatoid arthritis, but anti-inflammatory in other disease settings.⁷⁰ On the other hand, the low molecular weight form appears to be anti-inflammatory.⁷⁰ Clinical studies with FGF21 analogues have mostly measured levels of high molecular weight adiponectin. In the context of MASH, low levels of adiponectin are known to be associated with insulin resistance, and with steatosis and necroinflammation of the liver. Raised levels of adiponectin are associated with increased responsiveness of adipose tissue to insulin and with whole body sensitivity to insulin. Whether this is a causal relationship remains controversial.⁹ In the liver, adiponectin appears to promote oxidation of fat and inhibition of *de novo* lipogenesis via AMPK- and PPAR α -dependent signalling pathways, and exogenous adiponectin has been shown to alleviate steatohepatitis, hepatomegaly and fibrosis in rodent models of MASH.⁷¹ Loss of adiponectin function increases the susceptibility of mice to acetaminophen-induced hepatotoxicity.⁷² Adiponectin also appears to reduce ethanol-induced hepatocyte apoptosis *in vitro*.⁷³

Adiponectin has also been reported to modulate immune-effector cell function. It suppresses lipopolysaccharide-induced activation of Kupffer cells *in vitro* via a pathway involving IL-10,⁷⁴ STAT-3, and heme oxygenase-1. It has been reported to shift differentiation of monocytes to the pro-repair, anti-inflammatory phenotype of M2-like macrophages and suppress pro-inflammatory M1-like macrophages,⁷⁵ with inhibition of MCP-1 expression and macrophage infiltration associated with diet-induced liver inflammation.⁷⁶ Adiponectin has also been reported to block activation of HSCs by leptin.⁷⁷

In summary, current knowledge is insufficient to enable estimation of the extent to which the therapeutic effects of FGF21 analogues are mediated by adiponectin, despite several FGF21 analogues being associated with substantial increases in high-molecular weight adiponectin in patients with type 2 diabetes and/or MASH.

FGF21 analogues

The main factor limiting the use of native FGF21 is its short circulating half-life (30 min to 2 hours).⁷⁸ This is partly due to renal clearance because of its moderate molecular size, but largely due to inactivation. FGF21 is susceptible to cleavage of the C-terminal domain by an endopeptidase, FAP (fibroblast activating protein), which removes the last 10 amino acids, drastically reducing binding to the coreceptor β -klotho.^{79,80} Notably, FAP is highly expressed in the livers of patients with MASH.^{81,82} As such, engineering of FGF21 analogues to increase plasma half-life has been a prerequisite to study the

potential benefits of FGF21 in various metabolic conditions such as dyslipidaemia, obesity, type 2 diabetes, and MASLD.

Comparative primary pharmacology-pharmacokinetics

Pharmacodynamics of FGF21 analogues

The potency of the various FGF21 analogues has been reported based on activation of intracellular signalling in mammalian cells engineered to express human β -klotho or FGFR1c and β -klotho. The *in vitro* potency of FGF21 analogues depends on levels of FGFR1c and β -klotho on the surface of the different cell lines. By comparing them with the potency of recombinant human FGF21 (rhFGF21) measured concurrently, the reported relative *in vitro* potency of FGF21 analogues appears to be comparable and in the low nanomolar range.^{83–85} The potency of efruxifermin and pegozafermin, as agonists of FGFR 2c and 3c^{83,85} relative to rhFGF21, also appears to be comparable.

The binding kinetics of efruxifermin (with its divalent FGF21 configuration) to recombinant human β -klotho have been compared to matching single FGF21 chain configurations. The divalent configuration is associated with 4–9-fold higher rate of association, and 30–40-fold slower rate of dissociation compared to single chain configurations. This translates into 150–300-fold higher binding affinity.⁸⁶

The different design strategies of pegylated, lipidated or Fc-fusion-based FGF21 analogues have extended half-life to varying degrees. Critical to measuring half-life in serum is ensuring the immunoassay measures only biologically active drug. In practice this means either the capture or detection antibody must be specific for an intact C-terminal FGF21. While this has been confirmed for the assays used to measure pegbelfermin⁸⁷ and efruxifermin,⁸⁸ it has not been confirmed for NN0499, pegozafermin and BOS-580. With this caveat in mind, the published half-lives for efruxifermin and pegozafermin are approximately 3 days,^{68,85} and about 5 days for NN0499,⁸⁹ supporting sustained pharmacodynamic activity during their QW dosing regimens in phase II studies. The half-life of pegbelfermin was reported to be about 1 day, reflecting a less effective design for preventing cleavage of the C-terminus, which would not have sustained pharmacodynamic activity during the QW interval used for phase II. For BOS-580, the half-life is reported to be 21 days,²¹ with dosing intervals of up to 1 month being evaluated currently.

A second pharmacokinetic characteristic potentially influencing the pharmacodynamic activity of FGF21 analogues is exposure in the liver and peripheral compartments. Reducing flux of fat in the liver depends not only on inhibiting *de novo* lipogenesis, but also on inhibiting adipose tissue lipolysis.⁹⁰ Adiponectin level in serum is a marker of activation of FGFR1c, the predominantly expressed receptor of FGF21 in adipose tissue,⁹¹ which is an indicator of peripheral exposure. The maximal increase for adiponectin over baseline with the pegylated FGF21 analogue pegbelfermin was reported to be about 40–50% after dosing 30 mg QD for 14 days.⁸⁷ When dosed at 10–40 mg QW for up to 48 weeks, the maximal increase ranged from 25% to 50%.¹⁷ The magnitude of these

increases was smaller than reported for Fc-FGF21 analogues: efruxifermin (90–100% for 50 mg QW)^{13,15} and BOS-580 (103%, 300 mg QM).²¹ The smaller increases in adiponectin for pegbelfermin compared to Fc-FGF21 analogues suggest less peripheral exposure. This may be ascribed to the concentration of pegbelfermin in the liver, a phenomenon observed with pegylated insulin.⁹² The reported increase in adiponectin elicited by pegozafermin at the highest dose evaluated, 30 mg QW, was approximately 30%,¹⁶ intermediate between pegbelfermin and the Fc-FGF21 analogues, suggesting some concentration in the liver, but less than for pegbelfermin.

Comparison of primary pharmacology and pharmacodynamics of FGF21 analogues with an FGF-19 analogue

One recombinant human FGF19 (rhFGF19) analogue, NGM282, has been extensively evaluated in MASH. Native human FGF19 signals through a coreceptor complex of β -klotho with one of FGFR1c, 2c, 3c or 4. Because somatic mutations in FGFR4 are associated with hepatocellular cancer, FGF19 was engineered to eliminate activation of the tumorigenic STAT3 signalling pathway downstream of FGFR4,^{93–95} while retaining signalling through FGFR4 which modulates bile acid metabolism. The resulting rhFGF19, aldafermin, appears to activate coreceptor complexes of FGFR1c or 4 with β -klotho. Its potency against coreceptor complexes with FGFR2c and 3c has not been reported. Aldafermin reduces insulin resistance and lowers blood glucose in preclinical models of type 2 diabetes, which has been ascribed to activation of FGFR1c.⁹⁶ However, these effects have not been observed in patients with type 2 diabetes.⁹⁵ Activation of the coreceptor complex of β -klotho with FGFR4 profoundly suppresses expression of cytochrome P450 7A1,⁹⁷ a key regulatory enzyme of the metabolism of cholesterol to bile acids. Consequently, although aldafermin dosed at up to 6 mg QD to patients with MASH reduced liver steatosis and resolved MASH,⁹⁸ it was associated with substantial increases in low-density lipoprotein-cholesterol (LDL-c). This contrasts with FGF21 analogues which do not activate β -klotho/FGFR4 in humans and resolve MASH without increasing LDL-c.^{13,16,21}

Evidence from clinical trials

Late development drug candidates

Efruxifermin (formerly AKR-001, AMG 876)

Efruxifermin is a long-acting Fc-FGF21 fusion protein being developed for MASH and fibrosis.

Efruxifermin substantially reduced liver fat content in clinical trials, as assessed by MRI-proton density fat fraction (MRI-PDFF), both in the phase IIa BALANCED study (80 patients with MASH [F1–F3]; 16-week treatment), and the phase IIb HARMONY study (128 patients with MASH [F2/F3]; 24-week treatment). In the phase IIa study, the least-square (LS) mean relative percent changes from baseline to week 12 in liver fat content were –63.2%, –70.9% and –72.3% in the 28 mg, 50 mg and 70 mg groups dosed once weekly, respectively, and –0.3% in the placebo group.¹⁵ The percentage of patients with normalisation of liver fat content ($\leq 5\%$) was significantly higher

with efruxifermin than with placebo: 25%, 53%, and 67% of patients treated by efruxifermin 28 mg, 50 mg and 70 mg, respectively, vs. 5% in the placebo group.¹⁵ In the subsequent phase IIb study, efruxifermin demonstrated a comparable effect on liver fat reduction (Fig. 2). Efruxifermin also significantly improved liver histology and met its histological endpoints in phase IIb by reference to paired baseline biopsies (Fig. 2).¹³

Both the phase IIa and IIb trials reported several additional benefits including improved lipid profiles and glycaemic control (Fig. 2) arising from increased sensitivity to insulin in patients with or without type 2 diabetes, as well as a trend to weight loss with 50 mg and unchanged weight with 28 mg efruxifermin.¹³

Efruxifermin (50 mg) has also been studied in 30 patients with MASH and biopsy-confirmed compensated cirrhosis.¹⁴ After 16 weeks of treatment, improvements in serum biomarkers of fibrosis (ELF and Pro-C3) and liver injury (FAST) were noted compared to placebo. This pattern was consistent with histopathological results for a subset of patients who consented to end of treatment biopsies. Fifty eight percent of those treated with efruxifermin (7 of 12) had either an improvement of fibrosis or resolution of MASH, vs. 0% in a subset of the placebo group (0 of 5).¹⁴ A phase IIb trial (SYMMETRY) is currently ongoing and aims to confirm the ability of efruxifermin to reduce fibrosis in patients with MASH and compensated cirrhosis (NCT05039450). Encouraging results from the 36-week paired liver biopsy data have been reported.¹²⁵

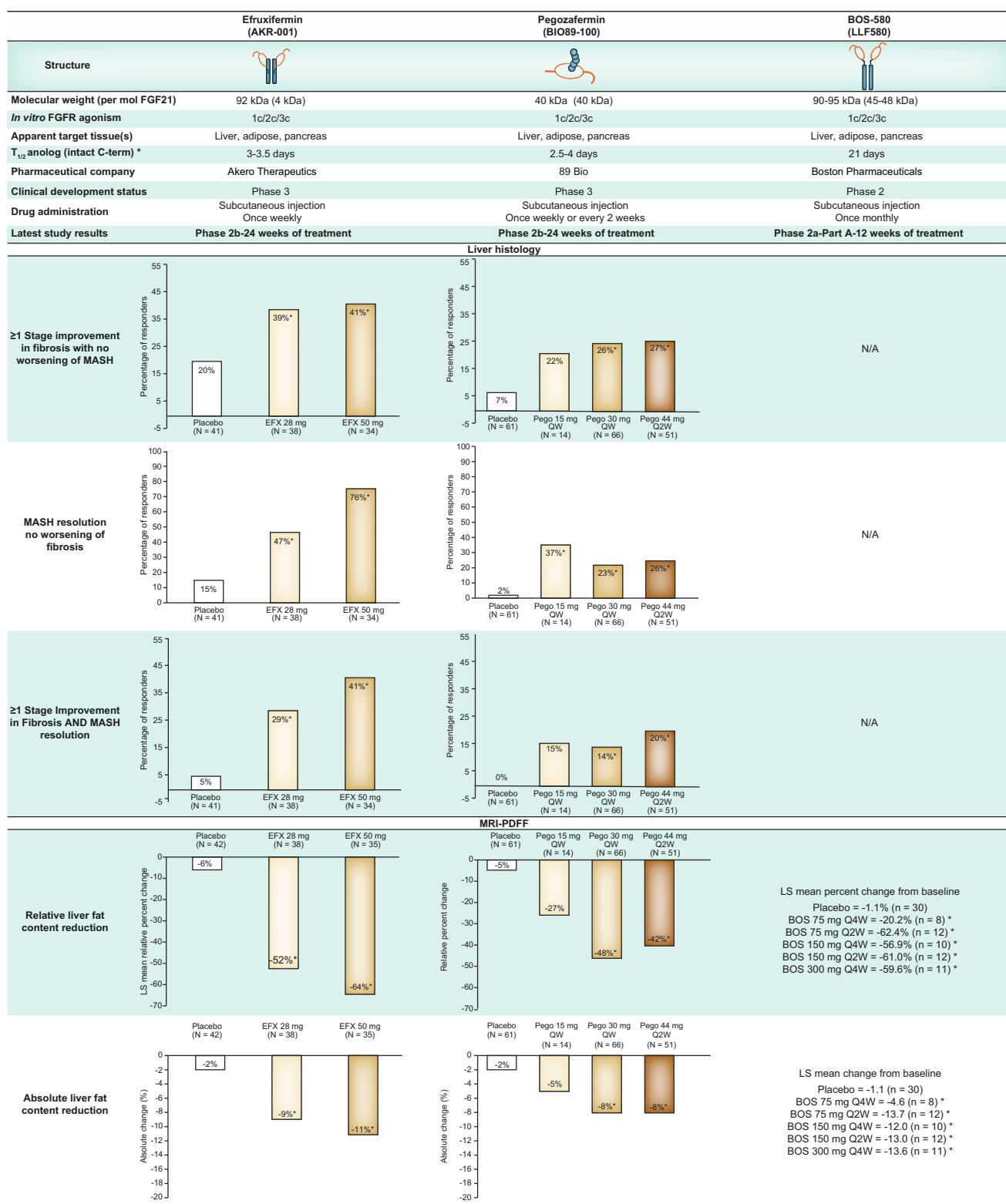
Subsequently a phase III program has been initiated. It includes three studies: 1) SYNCHRONY-HISTOLOGY to evaluate two doses of efruxifermin (28 mg and 50 mg) based on the primary endpoint of ≥ 1 stage improvement in fibrosis AND resolution of MASH in patients with MASH and fibrosis stage F2 or F3; 2) SYNCHRONY-REAL-WORLD to assess the safety and tolerability of efruxifermin based on non-invasive biomarkers in patients with MASLD/MASH; 3) SYNCHRONY-OUTCOMES to assess the progression to liver-related and all cause outcomes in patients with compensated cirrhosis due to MASH.

Pegozafermin (formerly BIO89-100)

Pegozafermin is a long-acting glyco-pegylated recombinant FGF21 analogue, which is also being developed for MASH and fibrosis.

Pegozafermin has been associated with significant liver fat content reduction, as assessed by MRI-PDFF, both in the phase Ib/IIa trial (81 patients with MASLD; 12-week treatment) and phase IIb trial (ENLIVEN, 222 patients with MASH [F2/F3]; 24-week treatment). In the phase Ib/IIa trial, the LS mean relative changes from baseline in liver fat content vs. placebo were –46.7% for 3 mg once weekly, –59.4% for 9 mg once weekly, –45.9 for 18 mg once weekly, –70.2% for 27 mg once weekly, –53.0% for 18 mg once every 2 weeks, and –60.2% for 36 mg once every 2 weeks.⁹⁹ In the subsequent phase IIb trial, pegozafermin demonstrated a consistent effect on liver fat reduction (Fig. 2).¹⁶ Pegozafermin also met its histological endpoints in its phase IIb study with paired baseline liver biopsies (Fig. 2).¹⁶

Given the positive phase IIb results, a registration phase III trial is underway to confirm the efficacy and safety of pegozafermin in patients with MASH and fibrosis.



	Efruxifermin (AKR-001)	Pegzofermin (BIO89-100)	BOS-580 (LLF580)
Structure			
Liver fat content normalization		N/A	N/A
Other imaging			
MRI-cT1	N/A	LS mean change from baseline versus placebo (n = 57) Pego 15 mg QW = -41 (n = 14) Pego 30 mg QW = -86 (n = 61) * Pego 44 mg Q2W = -64 (n = 49) *	N/A
Other noninvasive parameters			
Body Weight (kg)	LS mean change from baseline versus placebo (n = 42) EFX 28 mg QW = +0.4 (n = 37) EFX 50 mg QW = -2.3 (n = 36)	LS mean change from baseline versus placebo (n = 61) Pego 15 mg QW = +0.6 (n = 14) Pego 30 mg QW = -0.6 (n = 66) Pego 44 mg Q2W = +0.9 (n = 51)	N/A
HbA1c	Subset of T2DM patients LS mean change from baseline versus placebo (n = 27) EFX 28 mg QW = -0.5 (n = 28) * EFX 50 mg QW = -0.6 (n = 27) *	Subset of patients with HbA1c ≥6.5% LS mean change from baseline Placebo = -0.0 (n = 61) Pego 15 mg QW = -0.2 (n = 14) Pego 30 mg QW = -0.5 (n = 66) * Pego 44 mg Q2W = -0.4 (n = 51)	LS mean change from baseline Placebo = 0.0 (n = 30) BOS 75 mg Q4W = -0.0 (n = 8) BOS 75 mg Q2W = 0.0 (n = 12) BOS 150 mg Q4W = -0.2 (n = 8) BOS 150 mg Q2W = -0.3 (n = 12) BOS 300 mg Q4W = -0.4 (n = 10)
TG	LS mean percent change from baseline versus placebo (n = 42) EFX 28 mg QW = -34% (n = 35) * EFX 50 mg QW = -37% (n = 35) *	Median percent change from baseline versus placebo (n = 61) Pego 15 mg QW = -5.3% (n = 14) Pego 30 mg QW = -21.7% (n = 66) * Pego 44 mg Q2W = -8.2% (n = 51)	LS mean percent change from baseline Placebo = +7.7% (n = 30) BOS 75 mg Q4W = -1% (n = 8) BOS 75 mg Q2W = -33.5% (n = 12) * BOS 150 mg Q4W = -4.1% (n = 10) BOS 150 mg Q2W = -14.8% (n = 12) * BOS 300 mg Q4W = -23.2% (n = 11) *
AST	LS mean change from baseline versus placebo (n = 42) EFX 28 mg QW = -37% (n = 37) * EFX 50 mg QW = -47% (n = 36) *	LS mean percent change from baseline versus placebo (n = 61) Pego 15 mg QW = -34% (n = 14) * Pego 30 mg QW = -35% (n = 66) * Pego 44 mg Q2W = -29% (n = 51) *	LS mean percent change from baseline Placebo = +14% (n = 30) BOS 75 mg Q4W = -16% (n = 8) BOS 75 mg Q2W = -25% (n = 12) * BOS 150 mg Q4W = -31% (n = 10) * BOS 150 mg Q2W = -42% (n = 12) * BOS 300 mg Q4W = -43% (n = 11) *
ALT	LS mean change from baseline versus placebo (n = 42) EFX 28 mg QW = -34% (n = 35) * EFX 50 mg QW = -43% (n = 35) *	LS mean percent change from baseline versus placebo (n = 61) Pego 15 mg QW = -33% (n = 14) * Pego 30 mg QW = -37% (n = 66) * Pego 44 mg Q2W = -27% (n = 51) *	LS mean percent change from baseline Placebo = +8% BOS 75 mg Q4W = -24% (n = 8) * BOS 75 mg Q2W = -33% (n = 12) * BOS 150 mg Q4W = -25% (n = 10) * BOS 150 mg Q2W = -46% (n = 12) * BOS 300 mg Q4W = -45% (n = 11) *
FibroScan LSM	LS mean change from baseline versus placebo (n = 42) EFX 28 mg QW = -1.9 (n = 38) * EFX 50 mg QW = -3.6 (n = 36) *	LS mean change from baseline versus placebo (n = 61) Pego 15 mg QW = -2.2 (n = 14) Pego 30 mg QW = -3.9 (n = 66) * Pego 44 mg Q2W = -3.1 (n = 51) *	N/A
Pro-C3	LS mean percent change from baseline versus placebo (n = 40) EFX 28 mg QW = -27.1% (n = 37) * EFX 50 mg QW = -27.4% (n = 35) *	LS mean percent change from baseline versus placebo (n = 61) Pego 15 mg QW = -11.8% (n = 14) Pego 30 mg QW = -24.5% (n = 66) * Pego 44 mg Q2W = -23.7% (n = 51) *	LS mean percent change from baseline Placebo = -5.2% (n = 29) BOS 75 mg Q4W = -15.8% (n = 7) BOS 75 mg Q2W = -18.8% (n = 12) * BOS 150 mg Q4W = -11.2% (n = 10) BOS 150 mg Q2W = -29.7% (n = 12) * BOS 300 mg Q4W = -31.0% (n = 10) *
ELF	LS mean change from baseline versus placebo (n = 42) EFX 28 mg QW = -0.7 (n = 37) * EFX 50 mg QW = -0.8 (n = 34) *	LS mean from baseline versus placebo (n = 61) Pego 15 mg QW = -0.5 (n = 14) * Pego 30 mg QW = -0.5 (n = 66) * Pego 44 mg Q2W = -0.5 (n = 51) *	N/A
Adiponectin	LS mean percent change from baseline versus placebo (n = 42) EFX 28 mg QW = +34.0% (n = 36) * EFX 50 mg QW = +80.8% (n = 33) *	LS mean percent change from baseline versus placebo (n = 61) Pego 15 mg QW = +27.8% (n = 14) Pego 30 mg QW = +3702% (n = 66) * Pego 44 mg Q2W = +34.8% (n = 51) *	N/A

*Statistically significant : p value <0.005

*Efruxifermin half life: based on immunoassay which only recognizes intact C-terminal, Pegzofermin half life: species being measured has not been disclosed, BOS-580 half life: species being measured has not been disclosed.

Fig. 2. Comparison of FGF21 analogues in advanced phase clinical trials. *Efruxifermin half-life: based on immunoassay which only recognises intact C-terminal, Pegzofermin half-life: species being measured has not been disclosed, BOS-580 half-life: species being measured has not been disclosed.

In parallel, 89-BIO is developing pegzofermin for the treatment of severe hypertriglyceridemia. In a phase II trial (ENTRIGUE) including 85 patients with serum TG levels between 500–2,000 mg/dl, pegzofermin led to statistically

significant reductions in the primary endpoint of relative change in TG from baseline compared to placebo. A phase III trial (ENTRUST) to confirm the efficacy of pegzofermin for TG reduction started in Q2 2023 (NCT05852431).

Early development drug candidates

BOS-580 (formerly LLF580)

BOS-580 is a long-acting Fc-FGF21 fusion protein, developed by Boston Pharmaceuticals. A longer half-life has been reported for BOS-580 than for other FGF-21 analogues, potentially enabling up to once monthly subcutaneous dosing.²¹ However, it is not clear if this half-life refers to intact C-termini of the divalent FGF21 chains.

BOS-580 significantly reduced liver fat content, as assessed by MRI-PDFF, both in the phase I (64 obese patients with mild hypertriglyceridemia – 12 weeks of treatment)²¹ and in part A of an ongoing phase IIa trial (75 patients with MASLD; 12-week treatment).

In the part A of the phase IIa trial, the LS mean relative changes from baseline in liver fat content were -20.2%, -62.4%, -56.9%, -61.0% and -59.6% in the groups of patients who received BOS-580 75 mg Q4W, 75 mg Q2W, 150 mg Q4W, 150 mg Q2W and 300 mg (Q4W), respectively, vs. -1.1% for placebo.

In both trials, BOS-580 showed several additional benefits compared to placebo including reductions in markers of liver injury and fibrosis.²¹

Part B of this phase IIa in patients with MASH and fibrosis stages F2/F3 is currently ongoing to evaluate the efficacy of BOS-580 on secondary histological endpoints after 24 weeks of treatment (NCT04880031).

NNC0194 0499

NM0194 0499 is a lipidated FGF21 analogue, developed by Novo Nordisk.

In a phase I trial (57 obese patients – 12 weeks of treatment), NN0194 0499 demonstrated benefits on serum lipid profiles after a 12-week double-blind treatment period.⁸⁹ Placebo-adjusted reductions in TGs for the cohorts dosed once weekly with 3, 9, 27 and 60 mg were: 33%, 52%, 61% and 48%, respectively. Corresponding placebo-adjusted reductions in LDL-c were: -7%, 1%, 13% and 20%, respectively.

Terminated drug candidates

LY2405319

As the first analogue to enter clinical development, LY2405319, developed by Lilly, was the starting point for development of FGF21-based drugs.¹⁰⁰ However, despite significant weight loss and favourable changes in serum lipid profiles, particularly a substantial decrease in TG, LY2405319 was discontinued, likely owing to the absence of a robust glycaemic effect and hypersensitivity reactions.¹⁰¹

PF-05231023 (CVX-343)

PF-05231023, developed by Pfizer, was the second clinical experience for FGF21 analogue therapeutics. Early clinical trials confirmed benefits of this therapeutic class on lipid profiles. However, the short half-life of the intact C-termini of the divalent FGF21 chains of PF-05231023, resulting in high doses and increases in blood pressure and heart rate, likely led to its discontinuation.^{56,102}

Pegbelfermin (BMS-986036)

The first FGF21 analogue for which results on MASLD/MASH were reported was pegbelfermin, discovered by Ambrx and developed by Bristol Myers Squibb.

A trend toward reduction of liver fat content was observed in both phase IIb trials after 48 weeks of treatment with once weekly pegbelfermin (FALCON 1, 197 patients with MASH (F3), and FALCON 2, 155 patients with MASH and compensated cirrhosis).^{17,103}

Pegbelfermin failed to demonstrate statistical significance on histology endpoints in both phase IIb trials. In FALCON 1, the percentage of patients with an improvement in fibrosis of at least one stage without worsening of MASH was 14% to 16.3%, for those treated with 10–40 mg once weekly vs. 8.2% for placebo. The percentage of patients with MASH resolution without worsening of fibrosis was 2% to 8.2% for those treated with 10–40 mg once weekly vs. 6.1% for placebo.¹⁷ In FALCON 2, the proportion of patients with cirrhosis who achieved a ≥ 1 stage improvement in fibrosis without MASH worsening was 24.3% to 28.2% for those treated with 10–40 mg once weekly, vs. 30.8% for placebo.¹⁰³

In both FALCON 1 and 2 trials, the peak pharmacodynamic response occurred within the first 2–3 months of treatment, as indicated by change from baseline for non-invasive markers of liver injury and fibrosis, and a marker of FGFR1c activation in adipose tissue, adiponectin. After plateauing, the pharmacodynamic responses waned such that by 48 weeks they were unchanged relative to baseline. This observation has prompted questions about the sustainability of responses to FGF21 analogues. Tachyphylaxis of response to pegbelfermin did not appear to be related to presence of anti-pegbelfermin antibodies. When dosed once weekly, the magnitude of peak response for non-invasive markers was markedly smaller for pegbelfermin compared to either efruxifermin or pegozafermin. This is probably attributable to less effective agonism of FGF21's receptors by pegbelfermin, since the pharmacokinetic half-life was approximately 1 day compared to about 3 days for efruxifermin and pegozafermin. Recently published data for time courses of non-invasive markers of liver injury and fibrosis, as well as adiponectin, appears to show no discernible waning of response over 24 weeks of treatment with efruxifermin.¹³ Nevertheless, the potential for tachyphylaxis of response to FGF21 analogues should continue to be evaluated in longer term studies (≥ 52 weeks) as part of the phase III programmes.

Potential for combination therapy

Glucagon-like peptide 1 receptor agonist

Glucagon-like peptide 1 (GLP1) is an incretin hormone secreted during the postprandial phase by intestinal enterocytes. It regulates plasma glucose levels by stimulating pancreatic insulin secretion in response to glucose, while simultaneously inhibiting glucagon secretion. Semaglutide is a subcutaneous GLP1 receptor agonist (GLP1-RA), developed by Novo Nordisk and approved for the treatment of type 2 diabetes and obesity, which also reduces risk of cardiovascular disease and slows progression of renal disease, frequent comorbidities of MASH. It is currently in clinical development for MASH. A phase IIb trial, including 320 patients with F1-F3 MASH dosed daily with 0.4 mg semaglutide subcutaneously for 72 weeks, achieved the primary endpoint of resolution of MASH with no worsening of fibrosis together with a positive impact on glycaemic control.¹⁰⁴ However, it did not significantly reduce hepatic fibrosis. Novo Nordisk initiated its phase III registration trial (ESSENCE -

NCT04822181) in 2021, and interim surrogate results are expected in Q2 2024.

Combining treatment with an FGF21-analogue and GLP1-RA holds promise since they appear to possess complementary mechanisms, *i.e.* an insulin sensitiser vs. a secretagogue, while FGF21 exerts direct effects on hepatocytes and immune-effector cells resident in the liver, including direct suppression of collagen deposition and alleviation of hepatocyte stress. In a murine model, a novel brain GLP1-R-liver FGF21 axis mediating the weight-lowering effect of GLP1-RAs in the presence of high-carbohydrate diets has recently been described.¹⁰⁵ In this animal study, the absence of liver FGF21 expression was associated with resistance to liraglutide-induced weight loss, despite food intake being reduced by an independent neuro-nally mediated activation of GLP1-R.¹⁰⁵

Several companies have started clinical trials to explore the combination of these two therapeutics. Akero Therapeutics initiated a new cohort of patients with MASH, fibrosis F1 to F3, and type 2 diabetes, in the ongoing SYMMETRY trial to assess the safety and efficacy of efruxifermin as an add-on to existing GLP1-RA treatment (SYMMETRY Cohort D).¹⁰⁶ After 12 weeks of dosing with 50 mg efruxifermin once weekly, as an add-on to GLP1-RA treatment (diabetes-related dosage), the LS mean relative percent change from baseline in liver fat content was -65% vs. -10% in the GLP1-RA alone group, such that fat content was normalised (*i.e.* ≤5%) in almost 88% of efruxifermin+GLP1-RA-treated patients, compared to 10% of patients receiving GLP1-RA alone. In addition, the non-invasive marker of fibrosis, ELF, was significantly reduced in the combination group compared to the GLP1-RA alone group, suggesting the potential of efruxifermin to accelerate regression of fibrosis when used in addition to a GLP1-RA in patients with MASH.

Novo Nordisk also initiated a 52-week phase IIb trial to assess the efficacy and safety of NNC0194 0499 as monotherapy or as combination therapy with semaglutide in patients with MASH and fibrosis stages F2 to F4 (NCT05016882).

Thyroid hormone receptor β agonist

In the setting of lipotoxicity, hepatic thyroid function, mediated by THR- β , is disrupted, impacting lipoprotein metabolism and fatty acid oxidation in the liver. Targeting hepatic THR- β selectively modulates metabolic pathways in the liver without cardiac side effects arising from agonism of THR- α . THR- β -dependent signalling regulates bile acid, triglyceride, and cholesterol metabolism, reducing total cholesterol in serum and liver fat content. Top line results for resmetirom (first FDA conditionally approved drug for MASH), an oral, liver-targeting, selective THR- β agonist developed by Madrigal Pharmaceuticals Inc for the treatment of MASH, have recently been published.¹⁰⁷ An interim analysis of this trial (NCT03900429), including 966 patients treated daily for 52 weeks, demonstrated that both doses of resmetirom significantly improved the dual primary surrogate histological endpoints after 52 weeks of treatment (MASH resolution and ≥ 2 -point NAS [NAFLD activity score] reduction with no worsening of fibrosis, or fibrosis improvement by ≥ 1 stage with no worsening of NAS).¹⁰⁷

It has been shown that thyroid hormone regulates hepatic expression of FGF21.¹⁰⁸ The activation of THR- β induces the secretion of FGF21 by hepatocytes. Considering that THR- β

Box 1. Potential alternative disease indications for FGF21 analogues.

Other liver diseases:

- Alcohol-related steatohepatitis
- Haemochromatosis

Pulmonary, renal and myocardial fibrosis

Acute pancreatitis

Addiction (alcohol, opiates)

Chronic kidney diseases

Lipodystrophies

Myopathies and neuropathies caused by inherited mutations of mitochondrial proteins

Retinopathy of prematurity

Traumatic brain injury

FGF21, fibroblast growth factor 21.

agonism is likely to stimulate hepatic glucose output, the combination with an FGF21 analogue would potentially increase peripheral insulin sensitivity, which would result in increased glucose uptake by adipose tissue for storage as fat, while simultaneously preventing its release via lipolysis.

Alternative disease indications for FGF21 analogues (Box 1)

The essential role of FGF21 in protecting cells against oxidative and ER stress and suppressing pro-apoptotic signalling, as well as enabling cells to adapt to changing metabolic demands by increasing capacity for mitochondrial oxidation and autophagy highlights the potential of FGF21 analogues to treat a number of diseases or disorders associated with these underlying cell stressors.

Oxidative stress is a pathogenic driver of diseases/inherited disorders of the retina with its very high demand for energy and near complete depletion of oxygen across the arteriovenous circulation. Administration of FGF21 has been reported to ameliorate retinopathy of prematurity, which is associated with inherited mutations in the rhodopsin gene and is caused by excessive exposure of the developing eye of premature infants to oxygen. Myopathies caused by inherited mutations of mitochondrial proteins are associated with chronic oxidative stress, and carriers of such genetic variants characteristically have elevated circulating levels of FGF21, consistent with its role in mitigating the effects of oxidative stress. Renal tubular epithelial cells are another cell type particularly susceptible to oxidative stress. A common single nucleotide polymorphism associated with gain of FGF21 function is associated with reduced risk of chronic kidney disease, and improved markers of renal function.⁶⁵ A number of other liver diseases are associated with chronic oxidative stress, including alcohol-related steatohepatitis¹⁰⁹ and haemochromatosis.¹¹⁰

Exocrine acinar cells of the pancreas synthesise and secrete digestive enzymes at very high rates. Consequently, the high demands for protein translation, folding and packaging in secretory vesicles render acinar cells susceptible to ER stress, potentially leading to a localised inflammatory response which may manifest as acute pancreatitis. Clinical studies have shown that systemic FGF21 levels increase during episodes of pancreatitis, following slightly behind peak levels of amylase

and lipase. This has been interpreted as FGF21 being released as a compensatory response. Studies in a mouse model of acute pancreatitis have shown that knockout of FGF21 exacerbates the inflammatory response and fibrosis, whereas restoration of FGF21 reduces the extent of inflammation and fibrosis. FGF21 appears to alleviate ER stress by improving secretion of digestive enzymes without altering the rates of transcription or translation. Rates of protein translation and processing are also high in hepatocytes. Diseases like alpha-1 antitrypsin deficiency in serum are associated with reduced secretion by hepatocytes, arising from mutations preventing normal folding of the protein during post-translational processing with formation of insoluble intracellular aggregates and chronic ER stress.

Like MASH, many of the diseases associated with chronic inflammation lead to the development of fibrosis, compromising the function of the affected tissue or organ. Fibrotic diseases that represent major healthcare burdens and a large negative impact on quality of life include pulmonary, renal and myocardial fibrosis. In animal models, these fibrotic diseases, induced by a variety of non-metabolic insults, are worsened by loss and reversed by gain of FGF21 function.

Another group of diseases associated with profound insulin resistance, for example lipodystrophies, represent another potential therapeutic target for the insulin-sensitising activity of FGF21 analogues.

Finally, based on animal models and human genotype-phenotype associations and clinical observations, a number of diseases or disorders of the nervous system may be therapeutic targets for FGF21 analogues, including those associated with impaired mitochondrial function, traumatic brain injury and addiction. FGF21 and its receptors FGFR1c and β -klotho are expressed in several regions of the central nervous system (CNS) including the cortex, hippocampus, substantia nigra, basolateral amygdala and striatum.¹¹¹ In addition, it has been demonstrated that native FGF21 can cross the blood-brain barrier in a non-saturable fashion and without alteration in rodents.¹¹² Consistent with this, levels of FGF21 in human cerebrospinal fluid correlate with those in serum.¹¹³ The chronic metabolic effects of FGF21 in mice appear to be mediated through the CNS as effects on body weight and glucose homeostasis are lost when the coreceptor β -klotho is not expressed in the CNS. Several studies in animal models have demonstrated the potential effect of FGF21 in improving cognitive function, protecting against traumatic brain injuries, or reducing reward (opiate or alcohol)-seeking behaviour. However, these studies did not determine whether FGF21 was acting directly within the CNS or whether it was eliciting a centrally mediated response to afferent signals.

In this respect, mature native FGF21 has a molecular weight of approximately 20 kDa. In contrast, most FGF21 analogues are substantially larger, ranging from about 50 kDa for pegylated analogues to 90 kDa for Fc-fusion constructs. The larger molecular size would be expected to reduce transfer across the

blood-brain barrier. One exception is the lipidated analogue NNC0194 0499, with a molecular weight of approximately 21 kDa that is similar to native FGF21. Measurements of CNS exposure in either preclinical species or humans have not been reported for any of the FGF21 analogues currently in clinical development. Although in a cross-study comparison, the incidence of gastrointestinal adverse events was higher for the smallest FGF21 analogue, NNC0194 0499, than for the larger pegylated FGF21 analogues and efruxifermin.

Several preclinical and observational clinical studies have proposed that FGF21 plays a protective role in the cardiovascular system by reducing myocardial cell apoptosis, oxidative stress⁴⁵ and ischaemia-reperfusion injury,¹¹⁴ as well as reducing myocardial hypertrophy and fibrosis associated with chronic hypertension induced by overactive angiotensin-II pathway, DOCA-salt diet, high fructose intake or vascular calcification. Despite this, a recently reported Mendelian randomisation study based on a common human single nucleotide polymorphism associated with gain of FGF21 function did not appear to influence risk of coronary artery disease or heart failure, although blood pressure was slightly reduced.⁶⁵

Safety of FGF-21 analogues (Table 1)

Considering the pleiotropic effects of FGF21 on various organs/tissues, the potential side effects of long-term administration have been a focus of clinical development. A potentially detrimental impact on bone turnover was raised as a result of an animal model where inhibition of osteoblastogenesis and stimulation of adipogenesis from bone marrow by FGF21 resulted in increased bone loss.¹¹⁵ However, this was not replicated in another murine model under similar experimental conditions.¹¹⁶ Inhibition of growth hormone-induced secretion of insulin growth factor-1 observed in mice overexpressing FGF21 in their livers has also been raised as another mechanism which could shift the balance of bone metabolism toward resorption. However, this was associated with substantially reduced growth resulting from the negative energy balance induced by FGF21 and has not been observed consistently in mice with elevated levels of FGF21. In this respect, serum levels of total or free insulin growth factor-1 appeared to be unchanged or increased in several clinical studies with FGF21 analogues dosed up to 24 weeks.^{16,20,21,68} Among FGF21 analogues evaluated in clinical studies up to 48 weeks, bone mineral density by dual-energy X-ray absorptiometry was not meaningfully changed.^{16,17} Moreover, in these studies the incidence of bone fractures was comparable to placebo.¹⁰³ On the other hand, serum markers of bone accretion (procollagen type 1 N-terminal propeptide [P1NP], osteocalcin) and resorption (carboxy-terminal crosslinked telopeptide of type 1 collagen [CTX-1]) present a somewhat confusing picture following treatment with FGF21 analogues. Twelve-weeks B0S-580 treatment was associated with reductions in mean levels of P1NP and osteocalcin in patients with MAFLD, but CTX-1 was unchanged.²¹ Mean levels of P1NP declined and

Table 1. Current safety/tolerability profile of FGF21 analogues.

Tolerability	Safety	Additional questions
Mild to moderate gastrointestinal disorders (diarrhoea [mainly loose stool], nausea, increased appetite and vomiting) Injection-site reactions	Occasional hypoglycaemia Blood pressure fluctuations Bone health HPA axis activation?	Long-term consequences of ADA development?

ADA, anti-drug antibody; FGF21, fibroblast growth factor 21.

CTX-1 increased after 24 weeks of efruxifermin treatment in patients with MASH, although they remained within reference ranges.¹³ Interpretation of changes in biomarkers of bone turnover is complicated by potential confounding factors. Levels of P1NP appear to increase with advanced fibrosis of the liver.¹¹⁷ Consistent with this, baseline levels of P1NP before being treated with efruxifermin correlated with levels of P3NP (procollagen type 3 N-terminal peptide) a marker of soft tissue fibrillar collagen deposition.¹³ This strong correlation persisted during treatment with efruxifermin, indicating that the decline in P1NP paralleled that in P3NP. Another potential confounder is the susceptibility of patients with MASH to bone loss. Post-menopausal females represent a substantial segment of the population with MASH. In addition, many patients are taking proton pump inhibitors, antidepressants¹¹⁸ and systemic glucocorticoids,¹¹⁹ all of which have been reported to be associated with increased risk of osteoporosis or bone fracture. In this respect, there was a substantial unfavourable imbalance in use of these medications across the groups treated with efruxifermin for 24 weeks relative to placebo, as there was in the use of vitamin D supplements.¹³ Data from longer term phase III clinical trials will be needed to draw a clearer picture of the potential impact of FGF21 analogues on bone health.

Another preclinical safety concern identified in rodent studies was a potential increase in tone of the hypothalamic-pituitary-adrenal and sympathoadrenal axes which could lead to an increase in circulating levels of corticotropin-releasing hormone, cortisol, vasopressin and sympathetic nerve activity. It should be noted that an important biological role of FGF21 in rodents appears to be to augment non-shivering thermogenesis by both stimulating sympathetic innervation of brown adipose tissue, and to potentiate the activation of adrenoceptors on adipose tissue. The magnitude of such a role for FGF21 in humans appears to be lower, since hyperhidrosis (sweating) has been reported infrequently in human studies with FGF21 analogues. With respect to markers of hypothalamic-pituitary-adrenal activity, urinary free or salivary cortisol was unchanged after treatment with BOS-580 and efruxifermin, respectively.^{15,21} Corresponding data has not been reported for pegbelfermin or pegozafermin. Blood pressure and heart rate have been reported to be unchanged after treatment with BOS-580, pegozafermin, pegbelfermin or efruxifermin.^{13,16,21}

The most frequent adverse events reported across trials of FGF21 analogues are gastrointestinal disorders, including mild to moderate diarrhoea and nausea. As this gastrointestinal effect has not been observed in preclinical studies in rodents or non-human primates, the underlying mechanism is still unknown. The generally mild to moderate tolerability observed with FGF21 analogues may affect patient adherence; however, comparably severe gastrointestinal side effects are observed in several widely prescribed therapies for type 2 diabetes including metformin and GLP-1RAs. In addition, the gastrointestinal side effects with FGF21 analogues tend to be transient, occurring mainly during the first weeks of treatment, and tend to resolve on-treatment with minimal intervention, only occasionally resulting in discontinuation of treatment among approximately 5% of patients with F2-F3 MASH.^{13,15,16}

Injection-site reactions, including pain, induration, hypersensitivity, rash and erythema of mainly mild to moderate severity have also been reported for all FGF21 analogues.

Lastly, based on pegbelfermin and efruxifermin dosed for up to 48 and 24 weeks, respectively, FGF21 analogues, whether pegylated or Fc-fusion proteins, appear to elicit development of anti-drug antibodies (ADAs) at an incidence of 70–90% after 24 to 48 weeks treatment.^{12,14–17,103,120} The ADAs appear to develop slowly and are generally of low titre based on assays which meet current regulatory requirements for sensitivity of detection. The ADAs frequently cross-react with endogenous FGF21, but their presence does not appear to adversely affect pharmacodynamic response.^{13,17} Despite the high level of immunogenicity induced by efruxifermin and pegozafermin, there was no discernible attenuation of associated drug responses or adverse events related to those ADAs. Further and longer studies are warranted to fully assess the safety profile of the FGF21 analogues and their associated antibodies.

Conclusions

Promising results from two phase IIb trials of FGF21 analogues in patients with F2-F3 MASH meeting liver histology endpoints have been reported, with indications of benefit in more advanced fibrosis (F4) in phase IIa. The significant effect on both MASH resolution and fibrosis improvement, if confirmed in sub-part H registration trials, confers a key differentiation point for this class of therapeutics considering the worldwide market. Indeed, unlike the FDA, the EMA requires the use of a coprimary endpoint (MASH resolution **AND** improvement in fibrosis). Since there is no FDA- or EMA-approved therapy for MASH yet, a strict control of comorbidities is recommended in addition to lifestyle changes. For the many patients with MASH and comorbid type 2 diabetes this has meant treatment with pioglitazone or GLP1-RAs, or GLP1-RAs at a higher dose for those with obesity.¹²¹ However, in practice the use of pioglitazone is limited because of safety concerns due to oedema/volume loading among a population in which heart failure as well as weight gain are prevalent. Conversely, adoption of GLP1-RA therapy is expanding quickly in the MASH population because of their well-established efficacy on type 2 diabetes, weight management and improved cardiovascular and renal outcomes (SUSTAIN-6, PIONEER-6);¹²² all prevalent comorbidities in this population. However, GLP1-RAs do not appear to induce significant regression of fibrosis despite inducing MASH resolution to a similar extent after 72 weeks as achieved by FGF21 analogues after 24 weeks. The results presented at EASL show an effect on fibrosis.

If the lack of effect of semaglutide on fibrosis is confirmed during the ongoing registration trial to extend its label to include the MASH indication, this would essentially limit its use to resolution of MASH in patients with F2-F3 fibrosis in the US. A claim to only resolve MASH would not meet the medical need for patients with advanced fibrosis or cirrhosis in the US, nor permit approval for F2-F4 fibrosis in Europe. There is also an unresolved concern about potential loss of muscle that could be problematic in patients with MASH, with advanced fibrosis or cirrhosis, who are at high risk of developing sarcopenia.¹²³ In this future scenario, should FGF21 analogues be approved, they would likely be adopted widely since they have been shown to offer additional benefits on top of GLP1-RAs. Specifically, they would likely accelerate resolution of MASH (vis a vis efficacy of 24-week treatment with efruxifermin appears

comparable to 72-week treatment with semaglutide), and could lead to regression of fibrosis in a substantial proportion of patients, given the additional reductions of liver injury and fibrosis evident with 12 weeks of efruxifermin treatment on top of GLP1-RAs.¹⁰⁶ Moreover, the insulin-sensitising action of FGF21 analogues would complement GLP1-RAs (as insulin secretagogues) to further improve glycaemic control and reduce dyslipidaemia, thereby establishing a whole body metabolic environment conducive to enabling the liver to regenerate and to restore function. Resmetirom, a once-daily, oral THR- β agonist, has shown efficacy for MASH resolution and fibrosis improvement in a large phase III registration trial and has recently been approved by the FDA as the first MASH therapy in the US.¹²⁴ Significant improvement in LDL-c was also noted. Several additional oral drug candidates remain in development, including lanifibranor (pan-PPAR agonist) and

denifanstat (FASN inhibitor). Both have shown promising phase IIb results. Given the heterogeneity of MASH pathogenesis, it is likely that combination therapy will be needed to maximise efficacy in resolving steatohepatitis and inducing regression of fibrosis in the majority of patients diagnosed with MASH (F2-F4). As such, for those F2-F3 patients at highest risk of progression to cirrhosis, addition of an FGF21 analogue to other therapies approved for the treatment of MASH in the future could increase response rates for resolution of steatohepatitis and regression of fibrosis, while concurrently improving the underlying metabolic dysregulation.

In conclusion, considering their pleiotropic effects, consistent with improved liver health and systemic metabolism, and a strong anti-fibrotic effect, FGF21 analogues are a promising emerging therapy and may become a treatment of choice in the potential future armamentarium of MASH treatments.

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Abbreviations

ADAs, anti-drug antibodies; CNS, central nervous system; CTX-1, carboxy-terminal crosslinked telopeptide of type 1 collagen; EMA, European Medicines Agency; ER, endoplasmic reticulum; FDA, Food and Drug Administration; FGF, fibroblast growth factor; FGFRs, fibroblast growth factor receptors; GLP1, glucagon-like peptide 1; GLP1-RAs, GLP1 receptor agonists; HA, hyaluronic acid; HSCs, hepatic stellate cells; LDL-c, low-density lipoprotein-cholesterol; LS, least-square; MASH, metabolic dysfunction-associated steatohepatitis; MASLD, metabolic dysfunction-associated steatotic liver disease; P1NP, procollagen type 1 N-terminal propeptide; P3NP, procollagen type 3 N-terminal peptide; rhFGF21, recombinant human FGF21; THR, thyroid hormone receptor.

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Conflict of interest

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Please refer to the accompanying ICMJE disclosure forms for further details.

Authors' contributions

Preparation of the manuscript: All authors. Critical review of the manuscript: All authors. All authors approved the final version of the manuscript.

Supplementary data

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